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Key Points:

- Arc lavas yield fluxes of 4.0×10^{10} mol N/yr, similar to estimates from volcanic arc gases, likely resulting in net mantle ingassing of N
- Nitrogen isotopes and N-He mixing models highlight that small contributions of sediment dominate volcanic arc N budgets
- Subducted sediment thickness correlates with N_2/He ratios, and likely controls arc N fluxes rather than slab parameters or thermal state

Supporting Information:

Supporting Information may be found in the online version of this article.

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Deep Nitrogen Fluxes and Sources Constrained by Arc Lava Phenocrysts

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Abstract Nitrogen (N) dominates Earth's atmosphere (78% N₂) but occurs in trace abundances in silicate minerals, making it a sensitive tracer of recycled surface materials into the mantle. The mechanisms controlling N transfer between terrestrial reservoirs remain uncertain because low N abundances in mineral-hosted fluid inclusions (FIs) are difficult to measure. Using new techniques, we analyzed N and He isotope compositions and abundances in olivine- and pyroxene-hosted FIs from arc volcanoes in Southern Chile, Cascadia, Central America, and the Southern Marianas. These measurements enable an estimate of the global flux of N outgassing from arcs (4.0×10^{10} mol/yr). This suggests that Earth is currently in a state of net N ingassing, with roughly half of subducted N returned to the mantle. Additionally, the N outgassing flux of individual arcs correlates with the thickness of subducting pelagic sediment, suggesting that N cycling in the modern solid Earth is largely controlled by sediment subduction.

Plain Language Summary Nitrogen (N) largely behaves like an inert gas, and so it is substantially more concentrated at Earth's surface than in Earth's deep interior. Over geologic time, N can be transported between the solid Earth and the surface, and its concentration can change in both of these settings. Volcanic gases transport N from the interior to the surface, while some surface N returns into the solid Earth via plate subduction. Here, we present measurements of N and helium (He) gas trapped within crystals in volcanic rocks to determine how much N is transported to the surface through volcanism associated with plate subduction. We find that the amount of N returning to the surface through volcanism is less than estimates of how much N is transported into the solid Earth, suggesting that, overall, N is being returned to the planet's deep interior. Additionally, we observe that the amount of oceanic sediment that is subducted correlates with the amount of N that comes out of volcanoes, making it the primary carrier of N into the solid Earth.

1. Introduction: The Deep Nitrogen Cycle

The composition of Earth's atmosphere and its interior is regulated by the balance of subduction inputs and outputs via passive degassing and volcanism (Bekaert et al., 2021; Fischer, 2008; Hacker, 2008; Halama & Bebout, 2021; Hilton et al., 2002; Kelemen & Manning, 2015; Plank & Manning, 2019). The detailed mechanisms controlling the solid Earth fluxes of volatile species to the atmosphere remain murky, however, even for the most abundant atmospheric compound, N₂. Some studies imply net N ingassing into the mantle via subducted slabs (Bekaert et al., 2021; Labidi et al., 2020), whereas others suggest near complete return of subducted N to the surface at individual arcs (Elkins et al., 2006; Fischer et al., 2002; Labidi et al., 2021). It is even less clear how the N mass balance may have changed throughout Earth history. Evidence for relatively stable atmospheric N₂ concentrations since the Archean (Marty & Dauphas, 2003; Marty et al., 2013) is at odds with an evolving geotherm which may favor progressively more ingassing through time (Labidi et al., 2021; Mallik et al., 2018).

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Given the key role of N in Earth's habitability, the processes governing solid Earth N cycling, particularly during subduction, remain an important topic in the Earth sciences.

To understand past and present-day N fluxes, we need to determine the N sources in subducted oceanic lithosphere and sediments (i.e., slabs) and the impact of slab thermal regime on the high-pressure stability of N-bearing phases, like phengite (Förster et al., 2019; Halama et al., 2017; Liu et al., 2019). Subducted sediments likely dominate the N budget in many arcs (Fischer et al., 2002; Kita et al., 1993), though altered oceanic crust (AOC) has been proposed as the primary source of recycled N in the Marianas (Mitchell et al., 2010), New Zealand (Epstein et al., 2021), and Costa Rica (Lee et al., 2017). Evidence from experiments (Jackson et al., 2021), thermodynamic calculations (Mikhail et al., 2017; Mikhail & Sverjensky, 2014), and exhumed metasediments with variable P-T-t paths (Bebout et al., 2013) show that N mobilization from a subducting slab into the overlying mantle wedge should be sensitive to the thermal regime of modern subduction zones (Jackson et al., 2021; Labidi et al., 2021; Mallik et al., 2018; Mitchell et al., 2010). The consensus view holds that cooler thermal gradients associated with rapid subduction of colder slabs should facilitate N recycling to the mantle. However, it has also been proposed that hotter subduction may allow for greater slab melting. These melts may react with peridotitic mantle to form micas that could retain slab-derived N (Förster et al., 2024). In this case, “hot” subduction could result in less arc volcano outgassing of N. The role of slab temperature on terrestrial N cycling has not yet been systematically investigated with erupted arc magmas.

2. Methods

Nitrogen gas (N_2) concentrations and $\delta^{15}N$ compositions of olivine- and pyroxene-hosted fluid inclusions (FIs) were measured on the Nu Noblesse Noble Gas Mass Spectrometer in the Barry Lab at Woods Hole Oceanographic Institution (WHOI). Between 0.5 and 1.0 g of mineral separates were loaded into custom crushers and baked out overnight under high vacuum. Samples were then crushed at $\sim 5,000$ psi on a hydraulic press to extract fluids from inclusions. Liberated gases were then processed on a N_2 purification line before being inlet into the mass spectrometer. The WHOI purification line, previously described in Barry et al. (2012) and Barry and Hilton (2016), was modeled after the design of Takahata et al. (1998). Earlier measurements on arc samples made on the Takahata line (Fischer et al., 2005) are therefore directly comparable.

A new cleanup method was used to reduce blanks by an order of magnitude from previous methods (Barry et al., 2012; Barry & Hilton, 2016). The improved mass resolving power of the Nu Noblesse compared to older static mass spectrometers (e.g., VG 4500) can discriminate between the CO and N_2 peaks at masses 28 and 29. Examples of mass peaks, different volatile species, data quality, and reproducibility are described by Bekaert et al. (2023). Previously, these peaks could not be separated and a CuO furnace was cycled at high temperatures to convert CO to CO_2 . However, using the furnace to eliminate CO interferences causes relatively high blanks and high backgrounds, which is problematic for low N_2 samples like arc mineral separates. Gas concentrations measured by Bekaert et al. (2023) were sufficiently higher than the line blanks, so these authors kept the traditional step of converting CO with furnace cycling. However, we eliminated the CuO furnace cycling step, which is no longer necessary because the CO interference is now better quantified with greater mass resolving power on the Nu Noblesse. This reduced line blanks (and backgrounds) by more than an order of magnitude, and sometimes up to two orders of magnitude. The greatly reduced backgrounds allowed us to characterize low N mineral separates that previously presented a significant analytical challenge.

A line blank was analyzed for each of the two sample chambers prior to crushing. Line blanks are typically less than 2% even for low N_2 samples. Condensable gases were frozen out onto Pyrex cold fingers at liquid nitrogen temperatures and then expanded into the line. When ample gas was present, the expanded gas was analyzed. The remaining gases were then reanalyzed twice, such that each sample was analyzed three times. Air standards of known and approximately matching signal size were processed and analyzed in the same manner, also in triplicate. Air standards were used for blank-corrected calibration curves and linearity corrections for both N_2 concentrations and $\delta^{15}N$ of unknowns.

A second 0.5–1.0 g set of mineral separates was crushed and analyzed for He-Ne-Ar abundances and isotope compositions on MAP 215-50 mass spectrometer at the Woods Hole Oceanographic Institute Isotope Geochemistry Facility. Detailed methods are provided in Kurz et al. (2005, 2009).

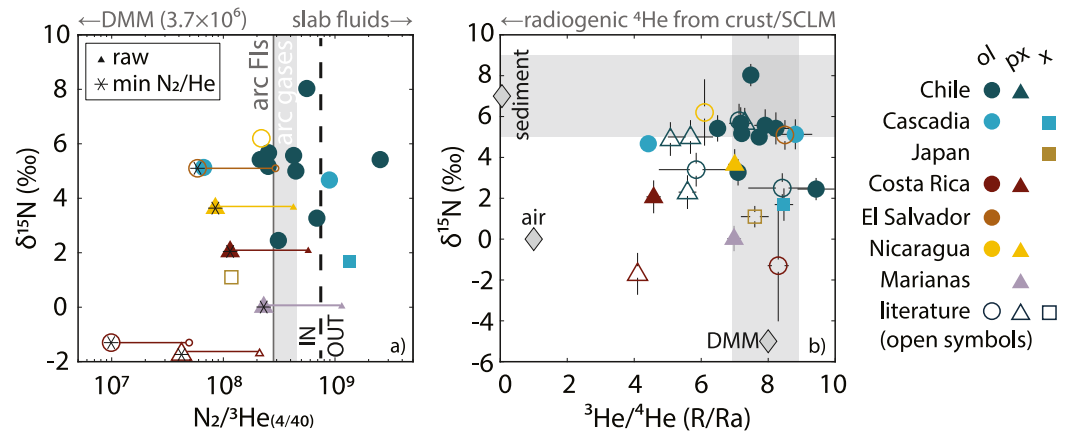


Figure 1. Nitrogen isotopes versus degassing-corrected $N_2/{}^3\text{He}_{(4/40)}$. For samples without complete He-N-Ar data, asterisks denote the “corrected” values used to calculate N fluxes. These use the possible $N_2/{}^3\text{He}_{(4/40)}$ from the raw data (small, connected symbols). The gray band gives the range of measured and flux-derived volcanic gas $N_2/{}^3\text{He}$ used for flux calculations in previous studies (Bekaert et al., 2021; Hilton et al., 2002). (Individual gas measurements from arcs extend above and below the x-axis range.) The dark gray line is our mean $N_2/{}^3\text{He}_{(4/40)}$. The dashed line is the threshold at which arc N fluxes are sufficient to cause net outgassing of N from the mantle. Ol = olivine phenocrysts, px = pyroxene phenocrysts, x = xenolith olivine. (b) $\delta^{15}\text{N}$ versus ${}^3\text{He}/{}^4\text{He}$ shows most fluid inclusion have mantle-like He isotopes and many also have sediment-like N isotopes. Lower ${}^3\text{He}/{}^4\text{He}$ indicate some amount of radiogenic crustal or SCLM contributions. Open symbols are literature N data (Fischer et al., 2005; Rouleau et al., 2018) and two filled samples use published He data (Shaw et al., 2006). Error bars represent 1 s.d.

3. Results

Samples measured in this study originate from arc volcanoes in Southern Chile, Cascadia, Central America, and the Southern Marianas. Abundances of ${}^3\text{He}$, ${}^4\text{He}$, ${}^{14}\text{N}$, ${}^{15}\text{N}$, and ${}^{40}\text{Ar}$ were determined by crushing 11 olivine phenocryst separates, 1 pyroxene phenocryst separate, and 1 olivine separate from a peridotite xenolith.

3.1. Gas Concentrations, Ratios, and Corrections

Raw N_2 concentrations in the mineral separates range from 3.5×10^{-6} to $8.5 \times 10^{-5} \text{ cm}^3\text{STP/g}$, and He ranges from 9.7×10^{-10} to $2.42 \times 10^{-8} \text{ cm}^3\text{STP/g}$ (Table S1 in Supporting Information S1). Relative elemental ratios of these gases are often more informative than concentrations, because of the inherent variation in the volume of FIs between samples. The N_2/He of the samples spans from 2.76×10^3 to 4.00×10^4 .

Solubility differences between the gases require a correction to assess the extent of degassing on the FI gas ratios (e.g., N_2/He , N_2/Ar , $N_2/{}^3\text{He}$), which are critical for determining fluid sources and estimating N fluxes, we apply a correction using ${}^4\text{He}/{}^{40}\text{Ar}^*$ values, where ${}^{40}\text{Ar}^* = {}^{36}\text{Ar}_{\text{sample}} \times ({}^{40}\text{Ar}/{}^{36}\text{Ar}_{\text{sample}} - {}^{40}\text{Ar}/{}^{36}\text{Ar}_{\text{air}})$, and ${}^{40}\text{Ar}/{}^{36}\text{Ar}_{\text{air}} = 298.56$ (Lee et al., 2006). Mantle production ratios of ${}^4\text{He}/{}^{40}\text{Ar}^*$ are approximately 2 ± 1 (Jambon et al., 1986; Marty & Zimmermann, 1999), but gases derived from the mantle may have higher or lower ${}^4\text{He}/{}^{40}\text{Ar}^*$ values due to the tenfold greater solubility of He in basaltic melts relative to Ar (Jambon et al., 1986; Marty, 1995) and the integrated degassing history of mantle melts.

The measured ${}^4\text{He}/{}^{40}\text{Ar}^*$ values in these samples ranged from 0.2 to 7.7. A mantle production value of ~ 2 can produce a minimum ${}^4\text{He}/{}^{40}\text{Ar}^*$ value of 0.2. Lower values (<2) are therefore considered to be derived from incipiently degassed fluids from a previously undegassed melt. Higher values (>2) reflect the effects of previous episodes of degassing that enriched the residual melt in He relative to Ar, as N_2 has a slightly higher solubility relative to Ar (Bekaert et al., 2023; Jambon et al., 1986; Marty, 1995). Applying solubility coefficients from Jambon et al. (1986) for He and Ar, and the N solubility from Marty (1995), we calculate the corresponding, degassing-corrected $N_2/{}^3\text{He}$ value (Figure 1a) using the following equation assuming a ${}^4\text{He}/{}^{40}\text{Ar}^*$ mantle production ratio of 2.

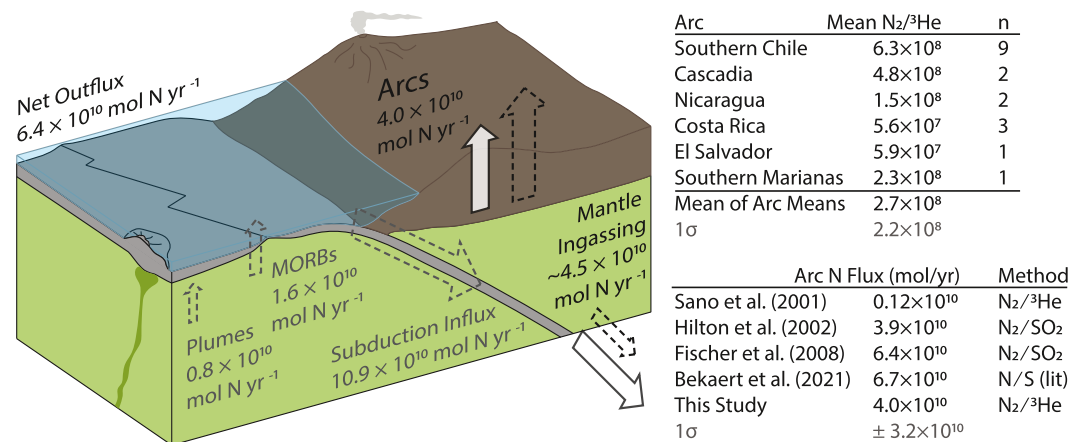


Figure 2. Schematic of N transport between Earth's surface and its interior. Fluxes are illustrated by arrows that are scaled by volume. Dashed arrows are the best estimates from the literature (Bekaert et al., 2021). White arrows are the fluxes calculated by this study. Mean arc segment N₂/³He_(4/40) values, previous arc N flux estimates from N₂/³He (Sano et al., 2001) and N₂/SO₂ of volcanic gases (Fischer, 2008; Hilton et al., 2002), and the ratio of arc N and S fluxes (Bekaert et al., 2021) are given.

$$\frac{N_2}{^3\text{He}_{(4/40)}} = \frac{N_2}{^3\text{He}_{\text{measured}}} \times \left(\frac{\frac{^4\text{He}}{^{40}\text{Ar}}_{\text{measured}}}{\frac{^4\text{He}}{^{40}\text{Ar}}_{\text{mantle}}} \right)^{\frac{(K_{\text{He}}^{-1} - K_{\text{N}_2}^{-1})}{(K_{\text{He}}^{-1} - K_{\text{Ar}}^{-1})}} \quad (1)$$

The solubility coefficients are $K_{\text{N}_2}^{-1} = 3.7 \times 10^{-12} \text{ mol g}^{-1} \text{ hPa}^{-1}$, $K_{\text{He}}^{-1} = 2.5 \times 10^{-11} \text{ mol g}^{-1} \text{ hPa}^{-1}$, and $K_{\text{Ar}}^{-1} = 2.6 \times 10^{-12} \text{ mol g}^{-1} \text{ hPa}^{-1}$. Corrected gas ratios are denoted with a _(4/40) subscript as in Equation 1.

Most N₂/He_(4/40) and N₂/³He_(4/40) data are within a factor of 2.5 of the uncorrected data. For samples without Ar data, we used the minimum ⁴He/⁴⁰Ar ratio of an incipiently degassed fluid (0.2) for the correction, which matches the lowest value in the data set. This yields minimum N₂/He_(4/40) and N₂/³He_(4/40) values that are a factor of 5 lower than the uncorrected N₂/He and N₂/³He results. The range of N₂/He_(4/40) is 734 to 29,000. Southern Chile and the Cascades generally have N₂/He_(4/40) in excess of 2,000 while Central America and the Southern Marianas yield lower values (Table S1 in Supporting Information S1). The range of N₂/³He_(4/40) is 6.7×10^7 to 2.5×10^9 , with generally higher values in Southern Chile and the Cascades and lower values in Central America and the Southern Marianas. The mean FI N₂/³He_(4/40) of our data and the literature data (Fischer et al., 2005) is consistent with median N₂/³He from ~700 volcanic arc gases (Hilton et al., 2002) and the ratio of N and ³He arc fluxes in a recent literature review (Bekaert et al., 2021), which illustrates that gas data and mineral separate data are directly comparable (Figures 1 and 2).

3.2. Nitrogen and Helium Isotopes

Nitrogen isotope abundances (¹⁴N and ¹⁵N) were collected for an additional 8 olivine and pyroxene phenocryst samples (Table S1 in Supporting Information S1). Nitrogen isotope compositions are given in delta notation (δ¹⁵N, in ‰) following Equation 2.

$$\delta^{15}\text{N}_{\text{sample}} = \left(\frac{^{15}\text{N}/^{14}\text{N}_{\text{sample}}}{^{15}\text{N}/^{14}\text{N}_{\text{air}}} - 1 \right) \times 1000 \quad (2)$$

The δ¹⁵N of phenocryst-hosted FIs (Figure 1a) range from −1.1‰ to +8.0‰ and are not strongly correlated with the N₂/³He_(4/40). However, our new data show that arc FIs record sediment-like δ¹⁵N and mantle-like ³He/⁴He of $8 \pm 1 \text{ R}_a$, where R_a is isotope ratio in air (Figure 1b; Graham, 2002). Crustal or subcontinental lithospheric mantle (SCLM) contamination of radiogenic ⁴He can explain ³He/⁴He < 7 R_a.

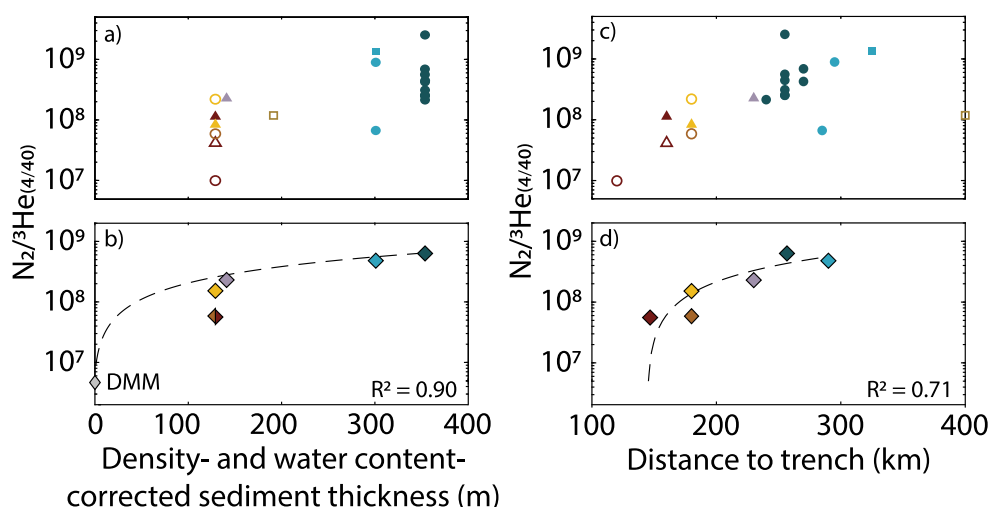


Figure 3. $N_2/{}^3\text{He}_{(4/40)}$ versus (a–b) sediment mass given as density- and water content-corrected sediment thicknesses (c–d) and the distance of the volcano from the trench. Individual measured data are shown separately (a–b) from the mean $N_2/{}^3\text{He}_{(4/40)}$ of phenocryst-hosted fluid inclusions for each arc segments (c–d; diamond symbols; see Figure 1 for additional symbol explanation). Dashed lines are linear best-fits with a y-intercept of 3.7×10^6 , corresponding to the DMM ratio (Labidi, 2022) and a minimum $N_2/{}^3\text{He}_{(4/40)}$ for an arc with no sediment subduction.

4. Discussion

4.1. Calculating Arc N Fluxes

A global N arc outflux estimate of $4.0 \pm 3.2 \times 10^{10}$ mol N/yr is calculated using the mean $N_2/{}^3\text{He}_{(4/40)}$ value (2.7×10^8 from all six arc segments analyzed) and an assumed annual arc outflux of 74 mol ${}^3\text{He}/\text{yr}$ (Bekaert et al., 2021), suggesting Earth is in a net ingassing state for N (Figures 1 and 2; Table S2 in Supporting Information S1). Considering average slab N influx estimates of 10.9×10^{10} mol/yr and mid-ocean ridge and plume N outflux estimates of 0.8×10^{10} and 1.6×10^{10} mol N/yr, respectively (Bekaert et al., 2021), our global arc N outflux estimate (4.0×10^{10} mol N/yr) would imply a net influx of N into the mantle of $\sim 4.5 \times 10^{10}$ mol/yr (Figure 2). This estimate is $\sim 40\%$ lower than recent estimates for arc N fluxes from global volatile flux literature reviews of 6.7×10^{10} mol/yr N derived from N_2/SO_2 (Fischer, 2008) and N and S fluxes (Bekaert et al., 2021) but comparable to an earlier estimate calculated from $N_2/{}^3\text{He}$ (Hilton et al., 2002). All three of these studies are based on air-corrected volcanic gas data. The value of $\sim 4.0 \times 10^{10}$ mol N/yr is a minimum N outgassing estimate because only minimum $N_2/{}^3\text{He}_{(4/40)}$ values could be determined for samples lacking complete N-He-Ar data sets (Fischer et al., 2005; Shaw et al., 2006), including some of our data. If we exclude these minimum $N_2/{}^3\text{He}_{(4/40)}$ values, we calculate a modestly higher arc N flux of 6.6×10^{10} mol N/yr from 3 arc segments. This modestly higher N flux reproduces those calculated from volcanic gas N_2/SO_2 measurements (Bekaert et al., 2021; Fischer, 2008) remarkably well, and corresponds to a global net ingassing flux of N of 1.9×10^{10} mol/yr. (For additional discussion about flux estimates from the literature, see Text S1 and Table S3 in Supporting Information S1).

We acknowledge that there are large uncertainties associated with both the global mean subduction fluxes of sedimentary N and the small number of arc FI data. However, we present this flux analysis to highlight that FI data yield similar estimates to previous N outgassing flux using volcanic gas data. We argue that convergence of N measurements in gas geochemistry data and in erupted materials bolsters confidence in global N flux estimates from arcs. Arc FIs represent a promising target for improving our understanding of N sources and the mechanisms that control N mobility in subduction zones, by allowing investigation at systems that are not vigorously degassing or through analysis of older materials to understand the temporal evolution of arcs.

4.2. Controls on N Fluxes—Slab Parameters or Subducted Sediment?

Variations in mean subducted thicknesses of pelagic and volcanoclastic sediments correlate with arc segment-averaged $N_2/{}^3\text{He}_{(4/40)}$ (Figures 3a and 3b; $R^2 = 0.90$), which suggests that arc N is primarily sourced from

sediments, consistent with previous findings (Fischer et al., 2002; Kita et al., 1993). Sediments in the Cascadia and Chilean trenches, which are thick relative to those offshore of Central America and the Marianas (Plank, 2014), likely impart a dominant control on $N_2/{}^3\text{He}_{(4/40)}$ and thus N flux. However, sediment packages for arcs in this study vary in lithology and N content, which explains some of the scatter in Figure 3a. Dominantly volcanoclastic sediments from the Marianas are lower in N concentration than the other arcs in this study (Plank, 2014), and carbonates and cherts with low N contents are subtracted from the total sediment thickness in our analysis (Table S4 in Supporting Information S1). Additionally, the thicknesses of subducted sediment packages can be highly variable, even within a single subduction zone (Text S2 in Supporting Information S1). For instance, in the Central Cascades, the thick Siletzia terrane impedes sediment subduction, and there is geophysical and geochemical evidence that this results in more sediment subduction and/or subcretion (i.e., underplating) in the Northern and Southern Cascades (Delph et al., 2021; Walowski et al., 2024). Similarly, the full sediment package offshore of Southern Chile is not subducted (Turner et al., 2016). Despite heterogeneity documented even within individual arcs, the correlation between our arc-averaged $N_2/{}^3\text{He}_{(4/40)}$ and the total thickness of subducted sediments highlights the clear role of sediments as a first order control on arc N fluxes. Given that factors including subduction erosion (Scholl & Von Huene, 2007) and underplating (Angiboust et al., 2022) can dramatically affect which materials and/or quantities of pelagic sediment that are transported to subarc depths, we find this correlation ($R^2 = 0.90$, Figure 3b) to be compelling.

We observe that slab thermal regime has a secondary effect on arc N fluxes. On an arc-by-arc basis, there is no observed trend between $N_2/{}^3\text{He}_{(4/40)}$ and major slab parameters (Figure S1 in Supporting Information S1; Text S3 in Supporting Information S1). Convergence rate (km/Ma), slab dip angle, and slab age (Ma) contribute to the slab thermal parameter (ϕ in km (Kirby et al., 1996; Syracuse & Abers, 2006; Syracuse et al., 2010)), and neither they nor ϕ correlate with N data. Over 2 orders of magnitude of variation are observed in $N_2/{}^3\text{He}_{(4/40)}$ at similar values of ϕ . The mean distance from the arc to the trench is the subduction parameter that best correlates with the FI $N_2/{}^3\text{He}_{(4/40)}$ data (Figures 3c and 3d). The arcs with the thicker sediment packages in this study are continental arcs with relatively young, hot, and buoyant subducting slabs. Continental margins accumulate more sediment and often have trenches that are further away from the arc (Scholl & Von Huene, 2007). While we do not argue for a causal relationship, it highlights the role of continental sediment production for subducted N fluxes, and ultimately, arc N outfluxes.

4.3. Nitrogen Sources and Subduction Parameters

Although sediments are the primary carrier of N to the mantle, they obscure the role of AOC in the deep N cycle, which has been shown to be important at sediment starved subduction zones (Epstein et al., 2021; H. Lee et al., 2017; Mitchell et al., 2010). The low concentrations of N in the mantle ($\sim 0.4 \mu\text{g/g}$; Cartigny & Marty, 2013) relative to recycled sediments ($100\text{--}2,000 \mu\text{g/g}$; Hebbeln et al., 2000; Li & Bebout, 2005; Martinez et al., 2006; Robinson et al., 2007; Sadofsky & Bebout, 2004) or AOC ($\sim 10 \mu\text{g/g}$; Busigny et al., 2019; Li & Li, 2022, 2023; Li et al., 2007) make small contributions from slab components readily apparent in FI $N_2/{}^3\text{He}_{(4/40)}$ (Figure 4a; Text S4 in Supporting Information S1). To trace N contributions from slab components, we invert the traditional $N_2/{}^3\text{He}$ ratio to account for the lack of He recycling from subducted slab components and allow their He/N_2 to be zero. Higher $\text{He}/N_{2(4/40)}$ ratios are indistinguishable from DMM, which in some cases (e.g., Costa Rica) is consistent with DMM-like N isotope values ($-5\text{‰} \delta^{15}\text{N}$), indicating minimal sedimentary inputs (Figure 4a). At lower FI $\text{He}/N_{2(4/40)}$ values, sediment and AOC contributions to arc N budgets can be distinguished using $\delta^{15}\text{N}$. Figure 4a shows that most FIs require low mass fractions (1%–10%) of sediment to explain the observed $\text{He}/N_{2(4/40)}$. Figure 4b shows subduction zone-specific mixing curves that use local sediment thicknesses (Plank, 2014), N concentrations, and $\delta^{15}\text{N}$ with constant contributions and compositions of AOC and mantle end-members (Table S4 in Supporting Information S1). These curves largely fit FI data, although Chilean samples record lower $\delta^{15}\text{N}$ than the mixing model predicts. Downcore sediment $\delta^{15}\text{N}$ values tend to decrease (Li & Bebout, 2005; Sadofsky & Bebout, 2004), so it may be that the mean uppermost sediment $\delta^{15}\text{N}$ (Hebbeln et al., 2000; Robinson et al., 2007) is overestimated. Alternatively, this could suggest either a greater contribution of AOC to the Chilean N budget or depletion in the $\delta^{15}\text{N}$ of Chilean arc magmas resulting from fluid-rock fractionation during slab devolatilization (Bebout et al., 2013; Bebout & Fogel, 1992). These data may also reflect interaction with other N sinks between the slab and arc magma reservoirs.

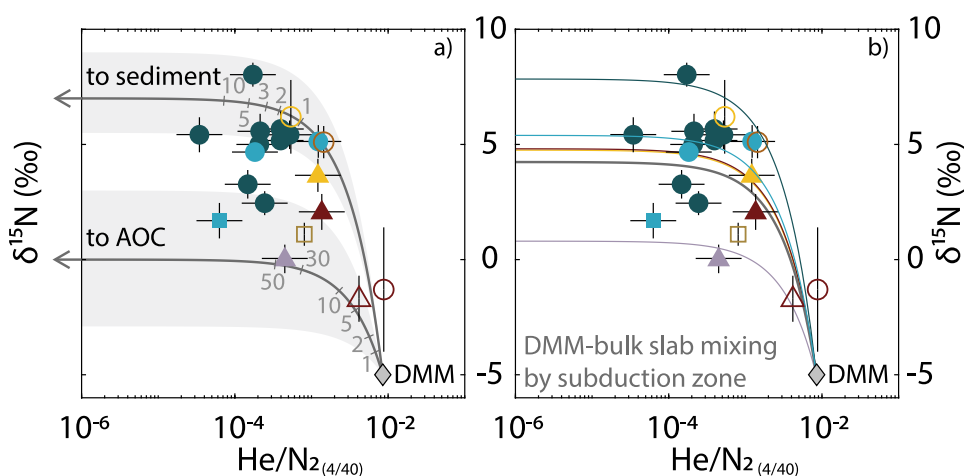


Figure 4. (a) DMM-sediment and DMM-AOC mixing models for $\delta^{15}\text{N}$ versus He/N_2 . Solid lines use the global mean N concentrations given by Bekaert et al. (2021). Shaded regions show possible mixtures given $\delta^{15}\text{N}$ uncertainty for sediment and altered oceanic crust (AOC). (b) Subduction zone-specific mixing models use subducted sediment masses, N concentrations, and $\delta^{15}\text{N}$ values for sediments, and $\delta^{15}\text{N}$ of AOC (Table S4 in Supporting Information S1). AOC mass and N content is held constant. Solid gray line is the global slab average model. He/N_2 errors are estimated to be $\pm 50\%$ and $\delta^{15}\text{N}$ errors are 1 s.d.

4.4. Additional N Sinks in Arcs

The processes controlling the fate of subducted N are complex and remain poorly constrained. Other sinks may exist that prevent N from being transported to the mantle or to the surface. For instance, exhumed high pressure metamorphosed slab lithologies (e.g., metasediments (Bebout et al., 2013; Bebout & Fogel, 1992; Sadofsky & Bebout, 2003)), metagabbros (Busigny et al., 2011), blueschists (Li et al., 2021), eclogites (Halama et al., 2010) contain much higher N concentrations and $\delta^{15}\text{N}$ than their protoliths, recording significant redistribution and storage of subducted N along the slab interface (Bebout & Fogel, 1992; Halama et al., 2010; Li et al., 2021). Similarly, forearcs can be important volatile sinks, as has been shown for other volatiles (e.g., carbon (Barry et al., 2019) and water (Hacker, 2008)). Nitrogen contents in forearc serpentinites and metaperidotites show these rocks are likewise capable of storing N (Cannaò et al., 2020; Pagé et al., 2018). The SCLM could also represent a significant long-term sink for volatiles (Gibson & McKenzie, 2023), including N (Barry & Broadley, 2021), that are neither outgassed to the surface nor ingassed to the mantle. Peridotite xenoliths from the Cascades and Japan (Fischer et al., 2005) have $\delta^{15}\text{N}$ values of 1.7 and 1.1‰, respectively (squares; Figures 1 and 4) and their high N_2 concentrations and $\delta^{15}\text{N}$ relative to DMM suggests metasomatism by slab fluids, which is supported by the presence of phlogopite in $\sim 5\%$ of Cascade mantle xenoliths (Ertan & Leeman, 1996). The extent to which these various environments represent additional sinks for subducted N is uncertain but represent a clear target for mass balance of N in future work.

5. Conclusions

Nitrogen and noble gas data from arc mineral separates ($\text{N}_2/{}^3\text{He}$) yield modern arc flux estimates (4.0×10^{10} mol N/yr) that are in good agreement with those derived from volcanic gas analyses (N_2/SO_2). This work represents the first N flux calculated from erupted materials in arcs, and builds toward a consensus on modern N cycling in the solid Earth. However, questions remain about the evolution of N on Earth through geologic time. For example, a hotter geotherm in the Archean and Paleoproterozoic is thought to have inhibited N recycling to the mantle, but the positive $\delta^{15}\text{N}$ observed in plumes globally (Cartigny & Marty, 2013; Marty & Dauphas, 2003) has long been regarded as evidence that some N in surface-derived materials was ingassed at that time. Our findings confirm previous work that demonstrated that sediment N influxes are intimately linked to arc N outfluxes (Fischer et al., 2002; Kita et al., 1993). The clear link between subducted sediment fluxes and arc N fluxes, and the lack of correlation between $\text{N}_2/{}^3\text{He}_{(4/40)}$ and slab parameters, suggest that a greater sediment flux alone may be responsible for the apparent net ingassing regime for N.

Data Availability Statement

All geochemical data, end member compositions, and sample information (Tables S1–S5 in Supporting Information S1) are archived on the Harvard Dataverse repository (<https://doi.org/10.7910/DVN/UJEGKF>) at Hudak et al. (2024).

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References

- Angiboust, S., Menant, A., Gerya, T., & Oncken, O. (2022). The rise and demise of deep accretionary wedges: A long-term field and numerical modeling perspective. *Geosphere*, 18(1), 69–103. <https://doi.org/10.1130/GES02392.1>
- Barry, P. H., & Broadley, M. W. (2021). Nitrogen and noble gases reveal a complex history of metasomatism in the Siberian lithospheric mantle. *Earth and Planetary Science Letters*, 556, 116707. <https://doi.org/10.1016/j.epsl.2020.116707>
- Barry, P. H., de Moor, J. M., Giovannelli, D., Schrenk, M., Hummer, D. R., Lopez, T., et al. (2019). Forearc carbon sink reduces long-term volatile recycling into the mantle. *Nature*, 568(7753), 487–492. <https://doi.org/10.1038/s41586-019-1131-5>
- Barry, P. H., & Hilton, D. R. (2016). Release of subducted sedimentary nitrogen throughout Earth's mantle. *Geochemical Perspectives Letters*, 2(2), 148–159. <https://doi.org/10.7185/geochemlet.1615>
- Barry, P. H., Hilton, D. R., Halldórsson, S. A., Hahm, D., & Marti, K. (2012). High precision nitrogen isotope measurements in oceanic basalts using a static triple collection noble gas mass spectrometer. *Geochemistry, Geophysics, Geosystems*, 13(1), 1–16. <https://doi.org/10.1029/2011GC003878>
- Bebout, G. E., Agard, P., Kobayashi, K., Moriguti, T., & Nakamura, E. (2013). Devolatilization history and trace element mobility in deeply subducted sedimentary rocks: Evidence from Western Alps HP/UHP suites. *Chemical Geology*, 342, 1–20. <https://doi.org/10.1016/j.chemgeo.2013.01.009>
- Bebout, G. E., & Fogel, M. L. (1992). Nitrogen-isotope compositions of metasedimentary rocks in the Catalina Schist, California: Implications for metamorphic devolatilization history. *Geochimica et Cosmochimica Acta*, 56(7), 2839–2849. [https://doi.org/10.1016/0016-7037\(92\)90363-N](https://doi.org/10.1016/0016-7037(92)90363-N)
- Bekaert, D. V., Barry, P. H., Curtice, J., Blusztajn, J., Hudak, M., Seltzer, A., et al. (2023). A carbon, nitrogen, and multi-isotope study of basalt glasses near 14°N on the Mid-Atlantic Ridge. Part A: Degassing processes. *Geochimica et Cosmochimica Acta*, 369, 160–178. <https://doi.org/10.1016/j.gca.2023.12.015>
- Bekaert, D. V., Turner, S. J., Broadley, M. W., Barnes, J. D., Halldórsson, S. A., Labidi, J., et al. (2021). Subduction-driven volatile recycling: A global mass balance. *Annual Review of Earth and Planetary Sciences*, 49(1), 37–70. <https://doi.org/10.1146/annurev-earth-071620-055024>
- Busigny, V., Cartigny, P., Laverne, C., Teagle, D., Bonifacie, M., & Agrinier, P. (2019). A re-assessment of the nitrogen geochemical behavior in upper oceanic crust from Hole 504B: Implications for subduction budget in Central America. *Earth and Planetary Science Letters*, 525, 115735. <https://doi.org/10.1016/j.epsl.2019.115735>
- Busigny, V., Cartigny, P., & Philippot, P. (2011). Nitrogen isotopes in ophiolitic metagabbros: A re-evaluation of modern nitrogen fluxes in subduction zones and implication for the early Earth atmosphere. *Geochimica et Cosmochimica Acta*, 75(23), 7502–7521. <https://doi.org/10.1016/j.gca.2011.09.049>
- Cannaò, E., Tiepolo, M., Bebout, G. E., & Scambelluri, M. (2020). Into the deep and beyond: Carbon and nitrogen subduction recycling in secondary peridotites. *Earth and Planetary Science Letters*, 543, 116328. <https://doi.org/10.1016/j.epsl.2020.116328>
- Cartigny, P., & Marty, B. (2013). Nitrogen isotopes and mantle geodynamics: The emergence of life and the atmosphere-crust-mantle connection. *Elements*, 9(5), 359–366. <https://doi.org/10.2113/gselements.9.5.359>
- Delph, J. R., Thomas, A. M., & Levander, A. (2021). Subcretionary tectonics: Linking variability in the expression of subduction along the Cascadia forearc. *Earth and Planetary Science Letters*, 556, 116724. <https://doi.org/10.1016/j.epsl.2020.116724>
- Elkins, L. J., Fischer, T. P., Hilton, D. R., Sharp, Z. D., McKnight, S., & Walker, J. (2006). Tracing nitrogen in volcanic and geothermal volatiles from the Nicaraguan volcanic front. *Geochimica et Cosmochimica Acta*, 70(20), 5215–5235. <https://doi.org/10.1016/j.gca.2006.07.024>
- Epstein, G. S., Bebout, G. E., Christenson, B. W., Sumino, H., Wada, I., Werner, C., & Hilton, D. R. (2021). Cycling of CO₂ and N₂ along the Hikurangi subduction margin, New Zealand: An integrated geological, theoretical, and isotopic approach. *Geochemistry, Geophysics, Geosystems*, 22(9), e2021GC009650. <https://doi.org/10.1029/2021GC009650>
- Ertan, I. E., & Leeman, W. P. (1996). Metasomatism of Cascades subarc mantle: Evidence from a rare phlogopite orthopyroxene xenolith. *Geology*, 24(5), 451–454. [https://doi.org/10.1130/0091-7613\(1996\)024<0451:mocsme>2.3.co;2](https://doi.org/10.1130/0091-7613(1996)024<0451:mocsme>2.3.co;2)
- Fischer, T. P. (2008). Fluxes of volatiles (H₂O, CO₂, N₂, Cl, F) from arc volcanoes. *Geochemical Journal*, 42(1), 21–38. <https://doi.org/10.2343/geochemj.42.21>
- Fischer, T. P., Hittton, D. R., Zimmer, M. M., Shaw, A. M., Sharp, Z. D., & Walker, J. A. (2002). Subduction and recycling of nitrogen along the Central American margin. *Science*, 297(5584), 1154–1157. <https://doi.org/10.1126/science.1073995>
- Fischer, T. P., Takahata, N., Sano, Y., Sumino, H., & Hilton, D. R. (2005). Nitrogen isotopes of the mantle: Insights from mineral separates. *Geophysical Research Letters*, 32(11), 1–5. <https://doi.org/10.1029/2005GL022792>
- Förster, M. W., Chen, C., Foley, S. F., Alard, O., & Yaxley, G. M. (2024). Fluid loss to the fore-arc controls the recycling efficiency of nitrogen in subduction zones. *Chemical Geology*, 650, 121985. <https://doi.org/10.1016/j.chemgeo.2024.121985>
- Förster, M. W., Foley, S. F., Alard, O., & Buhre, S. (2019). Partitioning of nitrogen during melting and recycling in subduction zones and the evolution of atmospheric nitrogen. *Chemical Geology*, 525(June), 334–342. <https://doi.org/10.1016/j.chemgeo.2019.07.042>
- Gibson, S. A., & McKenzie, D. (2023). On the role of Earth's lithospheric mantle in global volatile cycles. *Earth and Planetary Science Letters*, 602, 117946. <https://doi.org/10.1016/j.epsl.2022.117946>
- Graham, D. W. (2002). Noble gas isotope geochemistry of mid-ocean ridge and ocean island basalts: Characterization of mantle source reservoirs. *Reviews in Mineralogy and Geochemistry*, 47(1), 247–317. <https://doi.org/10.2138/rmg.2002.47.8>
- Hacker, B. R. (2008). H₂O subduction beyond arcs. *Geochemistry, Geophysics, Geosystems*, 9(3), Q03001. <https://doi.org/10.1029/2007GC001707>
- Halama, R., & Bebout, G. (2021). Earth's nitrogen and carbon cycles. *Space Science Reviews*, 217(3), 1–12. <https://doi.org/10.1007/s11214-021-00826-7>
- Halama, R., Bebout, G. E., John, T., & Schenk, V. (2010). Nitrogen recycling in subducted oceanic lithosphere: The record in high- and ultrahigh-pressure metabasaltic rocks. *Geochimica et Cosmochimica Acta*, 74(5), 1636–1652. <https://doi.org/10.1016/j.gca.2009.12.003>
- Halama, R., Bebout, G. E., Marschall, H. R., & John, T. (2017). Fluid-induced breakdown of white mica controls nitrogen transfer during fluid–rock interaction in subduction zones. *International Geology Review*, 59(5–6), 702–720. <https://doi.org/10.1080/00206814.2016.1233834>

- Hebbeln, D., Marchant, M., Freudenthal, T., & Wefer, G. (2000). Surface sediment distribution along the Chilean continental slope related to upwelling and productivity. *Marine Geology*, 164(3–4), 119–137. [https://doi.org/10.1016/S0025-3227\(99\)00129-2](https://doi.org/10.1016/S0025-3227(99)00129-2)
- Hilton, D. R., Fischer, T. P., & Marty, B. (2002). Noble gases and volatile recycling at subduction zones. *Reviews in Mineralogy and Geochemistry*, 47(1), 319–370. <https://doi.org/10.2138/rmg.2002.47.9>
- Hudak, M. R., Barry, P. H., Bekaert, D. V., Turner, S. J., Broadley, M. W., Walowski, K., et al. (2024). Data for “Deep Nitrogen Fluxes and Sources Constrained by Arc Lava Phenocrysts. *Harvard Dataverse*, V3. <https://doi.org/10.7910/DVN/UJEGKF>
- Jackson, C. R. M., Cottrell, E., & Andrews, B. (2021). Warm and oxidizing slabs limit ingassing efficiency of nitrogen to the mantle. *Earth and Planetary Science Letters*, 553, 116615. <https://doi.org/10.1016/j.epsl.2020.116615>
- Jambon, A., Weber, H., & Braun, O. (1986). Solubility of He, Ne, Ar, Kr and Xe in a basalt melt in the range 1250–1600 C. Geochemical implications. *Geochimica et Cosmochimica Acta*, 50(3), 401–408. [https://doi.org/10.1016/0016-7037\(86\)90193-6](https://doi.org/10.1016/0016-7037(86)90193-6)
- Kelemen, P. B., & Manning, C. E. (2015). Reevaluating carbon fluxes in subduction zones, what goes down, mostly comes up. *Proceedings of the National Academy of Sciences of the United States of America*, 112(30), E3997–E4006. <https://doi.org/10.1073/pnas.1507889112>
- Kirby, S. H., Stein, S., Okal, E. A., & Rubie, D. C. (1996). Metastable mantle phase transformations and deep earthquakes in subducting oceanic lithosphere. *Reviews of Geophysics*, 34(2), 261–306. <https://doi.org/10.1029/96rg01050>
- Kita, I., Nitta, K., Nagao, K., Taguchi, S., & Koga, A. (1993). Difference in N₂/Ar ratio of magmatic gases from northeast and southwest Japan: New evidence for different states of plate subduction. *Geology*, 21(5), 391. [https://doi.org/10.1130/0091-7613\(1993\)021<0391:DINARO>2.3.CO;2](https://doi.org/10.1130/0091-7613(1993)021<0391:DINARO>2.3.CO;2)
- Kurz, M. D., Curtice, J., Fornari, D., Geist, D., & Moreira, M. (2009). Primitive neon from the center of the Galápagos hotspot. *Earth and Planetary Science Letters*, 286(1–2), 23–34. <https://doi.org/10.1016/j.epsl.2009.06.008>
- Kurz, M. D., Moreira, M., Curtice, J., Lott, D. E., Mahoney, J. J., & Sinton, J. M. (2005). Correlated helium, neon, and melt production on the super-fast spreading East Pacific Rise near 17°S. *Earth and Planetary Science Letters*, 232(1–2), 125–142. <https://doi.org/10.1016/j.epsl.2005.01.005>
- Labidi, J. (2022). The origin of nitrogen in Earth’s mantle: Constraints from basalts ¹⁵N/¹⁴N and N₂/³He ratios. *Chemical Geology*, 597, 120780. <https://doi.org/10.1016/j.chemgeo.2022.120780>
- Labidi, J., Barry, P. H., Bekaert, D. V., Broadley, M. W., Marty, B., Giunta, T., et al. (2020). Hydrothermal ¹⁵N/¹⁴N abundances constrain the origins of mantle nitrogen. *Nature*, 580(7803), 367–371. <https://doi.org/10.1038/s41586-020-2173-4>
- Labidi, J., Young, E. D., Fischer, T. P., Barry, P. H., Ballentine, C. J., & de Moor, J. M. (2021). Recycling of nitrogen and light noble gases in the Central American subduction zone: Constraints from ¹⁵N/¹⁴N. *Earth and Planetary Science Letters*, 571, 117112. <https://doi.org/10.1016/j.epsl.2021.117112>
- Lee, H., Fischer, T. P., De Moor, J. M., Sharp, Z. D., Takahata, N., & Sano, Y. (2017). Nitrogen recycling at the Costa Rican subduction zone: The role of incoming plate structure. *Scientific Reports*, 7(1), 1–10. <https://doi.org/10.1038/s41598-017-14287-y>
- Lee, J.-Y., Marti, K., Severinghaus, J. P., Kawamura, K., Yoo, H.-S., Lee, J. B., & Kim, J. S. (2006). A redetermination of the isotopic abundances of atmospheric Ar. *Geochimica et Cosmochimica Acta*, 70(17), 4507–4512. <https://doi.org/10.1016/j.gca.2006.06.1563>
- Li, K., Li, G. Y., Du, Y. F., Han, W., Zhang, J., Chen, L. H., et al. (2021). Intralab remobilization of nitrogen during early subduction facilitates deep nitrogen recycling: Insights from the blueschists in the Heilongjiang Complex in NE China. *Chemical Geology*, 583(April), 120474. <https://doi.org/10.1016/j.chemgeo.2021.120474>
- Li, K., & Li, L. (2022). Nitrogen enrichments in sheeted dikes and gabbros from DSDP/ODP/IODP Hole 504B and 1256D: Insights into nitrogen recycling in Central America and global subduction zones. *Geochimica et Cosmochimica Acta*, 335, 197–210. <https://doi.org/10.1016/j.gca.2022.08.036>
- Li, K., & Li, L. (2023). Nitrogen enrichment in the altered upper oceanic crust: A new perspective on constraining the global subducting nitrogen budget and implications for subduction-zone nitrogen recycling. *Earth and Planetary Science Letters*, 602, 117960. <https://doi.org/10.1016/j.epsl.2022.117960>
- Li, L., & Bebout, G. E. (2005). Carbon and nitrogen geochemistry of sediments in the Central American convergent margin: Insights regarding subduction input fluxes, diagenesis, and paleoproductivity. *Journal of Geophysical Research*, 110(11), 1–17. <https://doi.org/10.1029/2004JB003276>
- Li, L., Bebout, G. E., & Idleman, B. D. (2007). Nitrogen concentration and δ¹⁵N of altered oceanic crust obtained on ODP Legs 129 and 185: Insights into alteration-related nitrogen enrichment and the nitrogen subduction budget. *Geochimica et Cosmochimica Acta*, 71(9), 2344–2360. <https://doi.org/10.1016/j.gca.2007.02.001>
- Liu, W., Yang, Y., Busigny, V., & Xia, Q.-K. (2019). Intimate link between ammonium loss of phengite and the deep Earth’s water cycle. *Earth and Planetary Science Letters*, 513, 95–102. <https://doi.org/10.1016/j.epsl.2019.02.022>
- Mallik, A., Li, Y., & Wiedenbeck, M. (2018). Nitrogen evolution within the Earth’s atmosphere-mantle system assessed by recycling in subduction zones. *Earth and Planetary Science Letters*, 482, 556–566. <https://doi.org/10.1016/j.epsl.2017.11.045>
- Martinez, P., Lamy, F., Robinson, R. R., Pichevin, L., & Billy, I. (2006). Atypical δ¹⁵N variations at the southern boundary of the East Pacific oxygen minimum zone over the last 50 ka. *Quaternary Science Reviews*, 25(21–22), 3017–3028. <https://doi.org/10.1016/j.quascirev.2006.04.009>
- Marty, B. (1995). Nitrogen content of the mantle inferred from N₂-Ar correlation in oceanic basalts. *Nature*, 377(6547), 326–329. <https://doi.org/10.1038/377326a0>
- Marty, B., & Dauphas, N. (2003). The nitrogen record for crust-mantle interaction and mantle convection from Archean to Present. *Earth and Planetary Science Letters*, 206(3–4), 397–410. [https://doi.org/10.1016/S0012-821X\(02\)01108-1](https://doi.org/10.1016/S0012-821X(02)01108-1)
- Marty, B., & Zimmermann, L. (1999). Volatiles (He, C, N, Ar) in mid-ocean ridge basalts: Assessment of shallow-level fractionation and characterization of source composition. *Geochimica et Cosmochimica Acta*, 63(21), 3619–3633. [https://doi.org/10.1016/S0016-7037\(99\)00169-6](https://doi.org/10.1016/S0016-7037(99)00169-6)
- Marty, B., Zimmermann, L., Pujol, M., Burgess, R., & Philippot, P. (2013). Nitrogen isotopic composition and density of the archaic atmosphere. *Science*, 342(6154), 101–104. <https://doi.org/10.1126/science.1240971>
- Mikhail, S., Barry, P. H., & Sverjensky, D. A. (2017). The relationship between mantle pH and the deep nitrogen cycle. *Geochimica et Cosmochimica Acta*, 209, 149–160. <https://doi.org/10.1016/j.gca.2017.04.007>
- Mikhail, S., & Sverjensky, D. A. (2014). Nitrogen speciation in upper mantle fluids and the origin of Earth’s nitrogen-rich atmosphere. *Nature Geoscience*, 7(11), 816–819. <https://doi.org/10.1038/ngeo2271>
- Mitchell, E. C., Fischer, T. P., Hilton, D. R., Hauri, E. H., Shaw, A. M., De Moor, J. M., et al. (2010). Nitrogen sources and recycling at subduction zones: Insights from the Izu-Bonin-Mariana arc. *Geochemistry, Geophysics, Geosystems*, 11(2), Q02X11. <https://doi.org/10.1029/2009GC002783>

- Pagé, L., Hattori, K., & Guillot, S. (2018). Mantle wedge serpentinites: A transient reservoir of halogens, boron, and nitrogen for the deeper mantle. *Geology*, 46(10), 883–886. <https://doi.org/10.1130/G45204.1>
- Plank, T. (2014). The chemical composition of subducting sediments. In *Treatise on geochemistry* (2nd ed., pp. 607–629). Elsevier.
- Plank, T., & Manning, C. E. (2019). Subducting carbon. *Nature*, 574(7778), 343–352. <https://doi.org/10.1038/s41586-019-1643-z>
- Robinson, R. S., Mix, A., & Martinez, P. (2007). Southern Ocean control on the extent of denitrification in the southeast Pacific over the last 70 ka. *Quaternary Science Reviews*, 26(1–2), 201–212. <https://doi.org/10.1016/j.quascirev.2006.08.005>
- Rouleau, E., Tardani, D., Vlastelic, I., Vinet, N., Sanchez, J., Sano, Y., & Takahata, N. (2018). Multi-element isotopic evolution of magmatic rocks from Cavihue-Copahue Volcanic Complex (Chile-Argentina): Involvement of mature slab recycled materials. *Chemical Geology*, 476, 370–388. <https://doi.org/10.1016/j.chemgeo.2017.11.035>
- Sadofsky, S. J., & Bebout, G. E. (2003). Record of forearc devolatilization in low-T, high-P/T metasedimentary suites: Significance for models of convergent margin chemical cycling. *Geochemistry, Geophysics, Geosystems*, 4(4), 1–29. <https://doi.org/10.1029/2002GC000412>
- Sadofsky, S. J., & Bebout, G. E. (2004). Nitrogen geochemistry of subducting sediments: New results from the Izu-Bonin-Mariana margin and insights regarding global nitrogen subduction. *Geochemistry, Geophysics, Geosystems*, 5(3), 1–23. <https://doi.org/10.1029/2003GC000543>
- Sano, Y., Takahata, N., Nishio, Y., Fischer, T. P., & Williams, S. N. (2001). Volcanic flux of nitrogen from the Earth. *Chemical Geology*, 171(3–4), 263–271. [https://doi.org/10.1016/S0009-2541\(00\)00252-7](https://doi.org/10.1016/S0009-2541(00)00252-7)
- Scholl, D. W., & Von Huene, R. (2007). Crustal recycling at modern subduction zones applied to the past—Issues of growth and preservation of continental basement crust, mantle geochemistry, and supercontinent reconstruction. In *Geological society of America memoirs* (Vol. 200, pp. 9–32). Geological Society of America. [https://doi.org/10.1130/2007.1200\(02\)](https://doi.org/10.1130/2007.1200(02))
- Shaw, A. M., Hilton, D. R., Fischer, T. P., Walker, J. A., & de Leeuw, G. A. M. (2006). Helium isotope variations in mineral separates from Costa Rica and Nicaragua: Assessing crustal contributions, timescale variations and diffusion-related mechanisms. *Chemical Geology*, 230(1–2), 124–139. <https://doi.org/10.1016/j.chemgeo.2005.12.003>
- Syracuse, E. M., & Abers, G. A. (2006). Global compilation of variations in slab depth beneath arc volcanoes and implications. *Geochemistry, Geophysics, Geosystems*, 7(5), Q05017. <https://doi.org/10.1029/2005GC001045>
- Syracuse, E. M., van Keken, P. E., Abers, G. A., Suetsugu, D., Bina, C., Inoue, T., et al. (2010). The global range of subduction zone thermal models. *Physics of the Earth and Planetary Interiors*, 183(1–2), 73–90. <https://doi.org/10.1016/j.pepi.2010.02.004>
- Takahata, N., Nishio, Y., Yoshida, N., & Sano, Y. (1998). Precise isotopic measurements of nitrogen at the sub-nanomole level. *Analytical Sciences*, 14(3), 485–491. <https://doi.org/10.2116/analsci.14.485>
- Turner, S. J., Langmuir, C. H., Katz, R. F., Dungan, M. A., & Escrig, S. (2016). Parental arc magma compositions dominantly controlled by mantle-wedge thermal structure. *Nature Geoscience*, 9(10), 772–776. <https://doi.org/10.1038/ngeo2788>
- Walowski, K. J., Wallace, P. J., DeBari, S. M., Wada, I., Shaw, S. D., & Rea, J. (2024). Disentangling the roles of subducted volatile contributions and mantle source heterogeneity in the production of magmas beneath the Washington Cascades. *Geochemistry, Geophysics, Geosystems*, 25(9), e2024GC011587. <https://doi.org/10.1029/2024GC011587>

References From the Supporting Information

- Carbotte, S. M., Boston, B., Han, S., Shuck, B., Beeson, J., Canales, J. P., et al. (2024). Subducting plate structure and megathrust morphology from deep seismic imaging linked to earthquake rupture segmentation at Cascadia. *Science Advances*, 10(23), ead13198. <https://doi.org/10.1126/sciadv.ad13198>
- Clift, P., & Vannucchi, P. (2004). Controls on tectonic accretion versus erosion in subduction zones: Implications for the origin and recycling of the continental crust. *Reviews of Geophysics*, 42(2), 2003RG000127. <https://doi.org/10.1029/2003RG000127>
- Contreras-Reyes, E. (2018). Structure and tectonics of the Chilean convergent margin from wide-angle seismic studies: A review. In A. Folguera, E. Contreras-Reyes, N. Heredia, A. Encinas, S. B. Iannelli, V. Oliveros, et al. (Eds.), *The evolution of the Chilean-argentinean andes* (pp. 3–29). Springer International Publishing. https://doi.org/10.1007/978-3-319-67774-3_1
- Craig, H., & Lupton, J. E. (1976). Primordial neon, helium, and hydrogen in oceanic basalts. *Earth and Planetary Science Letters*, 31(3), 369–385. [https://doi.org/10.1016/0012-821X\(76\)90118-7](https://doi.org/10.1016/0012-821X(76)90118-7)
- Davis, E. E., Fisher, A. T., & Firth, J. V. (Eds.) (1997). *Proceedings of the ocean drilling program, 168 initial reports* (Vol. 168). Ocean Drilling Program. <https://doi.org/10.2973/odp.proc.ir.168.1997>
- Han, S., Bangs, N. L., Carbotte, S. M., Saffer, D. M., & Gibson, J. C. (2017). Links between sediment consolidation and Cascadia megathrust slip behaviour. *Nature Geoscience*, 10(12), 954–959. <https://doi.org/10.1038/s41561-017-0007-2>
- Lyle, M., Koizumi, I., & Richter, C. (Eds.) (1997). *Proceedings of the ocean drilling program, 167 initial reports* (Vol. 167). Ocean Drilling Program. <https://doi.org/10.2973/odp.proc.ir.167.1997>
- Morris, J. D., Villinger, H. W., & Klaus, A. (Eds.) (2006). *Proceedings of the ocean drilling program, 205 scientific results* (Vol. 205). Ocean Drilling Program. <https://doi.org/10.2973/odp.proc.sr.205.2006>
- Mullen, E. K., Weis, D., Marsh, N. B., & Martindale, M. (2017). Primitive arc magma diversity: New geochemical insights in the Cascade Arc. *Chemical Geology*, 448, 43–70. <https://doi.org/10.1016/j.chemgeo.2016.11.006>
- Taira, A., Saito, S., Aoike, K., Morita, S., Tokuyama, H., Suyehiro, K., et al. (1998). Nature and growth rate of the Northern Izu–Bonin (Ogasawara) arc crust and their implications for continental crust formation. *Island Arc*, 7(3), 395–407. <https://doi.org/10.1111/j.1440-1738.1998.00198.x>
- Torgersen, T. (1989). Terrestrial helium degassing fluxes and the atmospheric helium budget: Implications with respect to the degassing processes of continental crust. *Chemical Geology: Isotope Geoscience section*, 79(1), 1–14. [https://doi.org/10.1016/0168-9622\(89\)90002-X](https://doi.org/10.1016/0168-9622(89)90002-X)
- Westbrook, G. K., Carson, B., & Musgrave, R. J. (Eds.) (1994). *Proceedings of the ocean drilling program, 146 Part 1 initial reports* (Vol. 146). Ocean Drilling Program. <https://doi.org/10.2973/odp.proc.ir.146-1.1994>